Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	
Project Name	EV Market Intelligence Dashboard
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Load the EV model dataset (model, manufacturer, country, category, battery, range, price, rating, 2025 sales) into the dashboard, with the option to replace the sample data with a real dataset.	High
2	Data Processing	Prepare the data for analysis: map fields, derive 2021-2025 sales history from category growth assumptions, and group records by category, battery, manufacturer and country.	High
3	Business Analysis	Calculate KPIs and metrics: annual sales, YoY growth, sales-weighted average range and rating, market share by category, manufacturer and battery, and the number of models and makers.	High
4	Business Insights & Analytical Outputs	Provide persona-based charts (category, manufacturer share, battery mix, range, ratings, sales trend, manufacturing country), KPI cards and an auto-generated insight summary.	High
5	User Interaction & Analysis	Let users filter by category, battery, manufacturing country and manufacturer, reset filters, switch persona tabs and sort the model explorer table.	High
6	Data Update & Reporting	Refresh KPIs, charts and the table instantly when filters or data change; support replacing the dataset and, in a later release, exporting charts and reports.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Clean, responsive layout with persona tabs, clear KPI cards and simple dropdown filters; works on desktop and mobile with light and dark themes.
NFR-2	**Security**	Runs in the browser with no login and no personal data; charting library loaded from a trusted CDN. Access

		control must be added before real or confidential data is used.
NFR-3	**Reliability**	Shows a clear message when no model matches the filters; all KPIs, charts and the table are computed from the same dataset so figures stay consistent.
NFR-4	**Performance**	KPIs, charts and the table re-render almost instantly after each filter change for the 40-model dataset; the dashboard is a single lightweight page.
NFR-5	**Availability**	Delivered as a static HTML page that any web host can serve and any modern browser can open at any time; there is no server or database to keep running.
NFR-6	**Scalability**	The dataset can grow beyond 40 models and add new categories, battery types or manufacturers without changing the layout; very large data would need a backend or BI platform.